

Employee Attrition Analysis

IBM HR Analytics Dataset | Descriptive Analytics Report

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Project Overview

This project analyzes employee attrition patterns using the IBM HR Analytics dataset, a widely used benchmark dataset containing 1,470 employee records across 35 variables. The goal was to apply a structured, end-to-end analytics workflow spanning data cleaning, exploratory analysis, SQL querying, and correlation analysis to identify the organizational and demographic factors most associated with voluntary employee departure.

The analysis was scoped to descriptive analytics, producing findings and recommendations directly applicable to HR decision-making without reliance on predictive modeling.

Dataset

Attribute	Detail
Source	IBM HR Analytics Employee Attrition & Performance (Kaggle)
Raw records	1,470 employees
Final shape after cleaning	1,470 rows x 32 columns
Columns removed	3 zero-variance columns (EmployeeCount, Over18, StandardHours)
Attrition rate	16.1% (237 of 1,470 employees)

Tools & Methods

- Python (pandas, seaborn, matplotlib) for data cleaning, EDA, and visualization
- SQLite via Jupyter Notebook for structured business question analysis across 6 queries
- Tableau for interactive dashboard design across two tabs: Overview and Risk Factors
- Correlation analysis with a tornado chart to rank numeric predictors of attrition

Key Findings

Six primary findings emerged from the combined EDA, SQL, and correlation phases:

Finding	Key Metric
Sales Representatives have the highest attrition rate	39.8% vs. 4.9% for Managers
Overtime is the strongest categorical attrition driver	30.5% leave vs. 10.4% for non-overtime employees
Attrition is concentrated in the first year	36% attrition rate in Year 1; drops sharply after Year 5
Employees who left earned significantly less	\$4,787/month vs. \$6,833 for retained employees (\$2,046 gap)
Distance from home is the top numeric predictor	21.4% attrition for far employees vs. 14.1% for nearby employees
Single employees leave at more than double the rate of divorced employees	25.5% vs. 10.1%

Analysis Phases

Phase 1 & 2 — Setup and Cleaning

Loaded the raw IBM CSV into a pandas DataFrame. Identified and removed three zero-variance columns that carried no analytical value (EmployeeCount, Over18, StandardHours). Verified no null values were present. Final dataset shape: 1,470 rows x 32 columns. Exported clean_attrition.csv for downstream use in SQL and Tableau.

Phase 3 — Exploratory Data Analysis

Generated five seaborn visualizations to surface initial attrition patterns:

- Attrition by Tenure — tenure distribution for employees who left vs. stayed, revealing a Year 1 spike
- Attrition by Job Role — horizontal bar chart showing Sales Representatives as the highest-risk role
- Attrition by Department — grouped bar chart comparing attrition across Sales, R&D, and HR
- Attrition by Overtime Status — bar chart showing a 3x attrition gap between overtime and non-overtime employees
- Income by Attrition Status — boxplot revealing a \$2,046/month median income gap

Phase 4 — SQL Analysis (SQLite)

Loaded `clean_attrition.csv` into a SQLite database within Jupyter Notebook and ran six structured queries:

- Q1: Overall attrition rate by count and percentage
- Q2: Attrition rate by Job Role, ranked descending
- Q3: Attrition rate by Department
- Q4: Attrition rate segmented by Overtime status
- Q5: Average monthly income segmented by attrition status
- Q6: Attrition rate segmented by distance from home (near/medium/far bins)

Phase 5 — Correlation Analysis

Computed point-biserial correlations between all numeric features and the binary attrition target variable. Visualized results as a tornado chart sorted by absolute correlation strength. Top positive predictors of attrition included `OverTime` (encoded), `DistanceFromHome`, and `NumCompaniesWorked`. Top negative predictors (associated with retention) included `TotalWorkingYears`, `JobLevel`, `Age`, and `MonthlyIncome`.

Tableau Dashboards

Two dashboards were built in Tableau using a consistent coral (#E15759) and gray (#BAB0AC) color system, with coral representing attrition and gray representing retention or comparison groups. All chart titles are insight-driven rather than descriptive.

- Overview Dashboard — attrition by job role, department, overtime status, and tenure. Audience: HR leadership and department managers.
- Risk Factors Dashboard — attrition by distance from home, marital status, and income distribution. Audience: HR analysts and workforce planning teams.

Dashboard link: [\[Tableau Public\]](#)

Recommendations

Based on the analysis, three targeted recommendations were developed:

1. Audit overtime distribution and compensation for Sales Representatives.

Sales Representatives leave at nearly 4x the rate of Managers, and overtime employees leave at 3x the rate of non-overtime employees. These factors likely compound in sales roles. Auditing workload distribution and benchmarking compensation against market rates would directly address the two strongest attrition drivers identified.

2. Strengthen Year 1 onboarding and retention touchpoints.

Attrition peaks at 36% in the first year and falls sharply after Year 5, indicating that early-tenure employees are the highest flight risk. Structured check-ins, mentorship pairings, and targeted engagement programs during the first 12 months would have the highest ROI for retention efforts.

3. Develop flexible work options for employees with long commutes.

Distance from home is the strongest numeric predictor of attrition in the correlation analysis, with far employees leaving at 21.4% vs. 14.1% for nearby employees. Remote or hybrid flexibility for employees outside a defined commute threshold could reduce this risk without compensation changes.

Limitations

- Dataset is simulated. The IBM HR Analytics dataset is a widely used fictional dataset and does not represent a real organization. Findings are appropriate for portfolio demonstration, not organizational decision-making.
- No temporal dimension. The dataset is a cross-sectional snapshot and does not capture changes over time or attrition cause (voluntary vs. involuntary).
- Correlation is not causation. Relationships identified in the correlation analysis reflect statistical association, not causal mechanisms.